360digrii

Data Analytics – SQL 01

Table: employees

emp_id	name	department	salary	age	city	joining_date
1	Ravi Kumar	HR	40000	28	Chennai	2021-06-15
2	Suresh Babu	IT	60000	32	Hyderabad	2020-03-10
3	Lakshmi Devi	Finance	55000	30	Bengaluru	2019-07-23
4	Arun Prasad	IT	70000	35	Chennai	2018-01-12
5	Meena Iyer	HR	45000	29	Kochi	2022-09-01
6	Karthik Reddy	IT	65000	31	Hyderabad	2021-11-19
7	Divya Nair	Finance	50000	27	Kochi	2020-05-30
8	Prakash Rao	IT	72000	36	Bengaluru	2017-08-14

Table: orders

order_id	emp_id	order_amount	order_date
101	1	5000	2023-01-10
102	2	7000	2023-02-15
103	3	6000	2023-03-20
104	4	8000	2023-04-25
105	5	4500	2023-05-05
106	6	7500	2023-06-18
107	7	5200	2023-07-22
108	8	9000	2023-08-30

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Section A: MCQs (15 Questions)

1. Which operator is used for pattern matching?

A) LIKE
B) =
C) IN
D) BETWEEN

2. What does WHERE salary > 50000 do?

A) Sorts rows
B) Groups rows
C) Filters rows
D) Joins tables

3. Which clause is used to filter grouped data?

A) HAVING
B) WHERE
C) ORDER BY
D) LIMIT

4. What is the result of salary BETWEEN 40000 AND 60000?

A) Only 40000
B) Excludes boundary values
C) Includes both 40000 and 60000
D) Only 60000

5. Which operator checks if a value exists in a list?

A) EXISTS
B) BETWEEN
C) LIKE
D) IN

6. Which function assigns rank without gaps?

A) ROW_NUMBER()
B) RANK()
C) NTILE()
D) DENSE_RANK()

7. What does ROW_NUMBER() do?

A) Assigns unique sequential numbers
B) Returns total rows
C) Groups data
D) Filters duplicates

8. Which clause is mandatory for window functions?

A) GROUP BY
B) HAVING
C) OVER()
D) LIMIT

9. What will WHERE city = 'Chennai' return?

A) All employees
B) Employees from Hyderabad
C) Only Chennai employees
D) NULL values

10. Which operator is used to check NULL values?

A) = NULL
B) NULL
C) IS NULL
D) LIKE NULL

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11. What does ORDER BY salary DESC do?

A) Ascending sort
B) Groups data
C) Filters data
D) Descending sort

12. Which window function is commonly used for Top-N per group?

A) AVG()
B) COUNT()
C) RANK()
D) SUM()

13. What does LIMIT 5 do?

A) Filters columns
B) Limits number of rows returned
C) Sorts data
D) Groups data

14. Which clause is executed first in SQL query processing?

A) SELECT
B) ORDER BY
C) GROUP BY
D) WHERE

15. What does LIKE '%Kumar' mean?

A) Starts with Kumar
B) Contains Kumar in middle
C) Exact match
D) Ends with Kumar

Section B: SQL Queries (10 Questions)

Write SQL queries for the following:

1.Select all employees with salary greater than 50,000.

2.Find employees from Chennai or Hyderabad.

3.Display employees whose name starts with 'K'.

4.Get employees aged between 28 and 35.

5.Find total salary by department.

6.Get top 3 highest-paid employees.

7.Assign row numbers to employees based on salary (highest first).

8.Rank employees within each department based on salary.

9.Find employees whose salary is above department average (use window function).

10.Get cumulative salary by department (running total).

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Section C: SQL Queries (10 Questions)

Q1 Get employee names with their department names

Part 1 (SELECT)
A. SELECT e.name, d.dept_name
B. SELECT d.dept_name
C. SELECT e.salary, d.dept_name

Part 2 (FROM + JOIN)
A. employees e LEFT JOIN departments d
B. employees e INNER JOIN departments d
C. departments d JOIN employees e

Part 3 (Condition)
A. ON e.id = d.id
B. ON d.id > 10
C. ON e.dept_id = d.id

Options:
A. A, B, C
B. B, C, A
C. C, A, B
D. A, C, B

Q2 Count employees in each department

Part 1
A. SELECT dept_id, COUNT()
B. SELECT name, COUNT()
C. SELECT COUNT(*)

Part 2
A. employees
B. departments
C. employees e

Part 3
A. GROUP BY name
B. GROUP BY dept_id
C. ORDER BY dept_id

Options:
A. B, A, B
B. A, A, B
C. C, B, C
D. A, C, A

Q3 Departments with more than 5 employees

Part 1
A. SELECT dept_id
B. SELECT dept_id, COUNT()
C. SELECT COUNT()

Part 2
A. employees
B. departments
C. employees e

Part 3
A. WHERE COUNT() > 5
B. HAVING dept_id > 5
C. GROUP BY dept_id HAVING COUNT() > 5

Options:
A. B, A, C
B. A, B, A
C. C, A, B
D. B, C, A

Q4 Employees with salary above average

Part 1
A. SELECT name
B. SELECT salary
C. SELECT emp_id

Part 2
A. employees
B. departments
C. salaries

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Part 3

A. WHERE salary < 50000

B. WHERE salary > (SELECT AVG(salary) FROM employees)

C. WHERE dept_id IN (1,2)

Options:

A. B, A, A

B. A, A, B

C. C, B, C

D. A, C, A

Q5 Employees in departments 1,2,3

Part 1

A. SELECT name

B. SELECT dept_id

C. SELECT salary

Part 2

A. employees

B. departments

C. employee_data

Part 3

A. WHERE dept_id > 5

B. WHERE dept_id IN (1,2,3)

C. WHERE dept_id NOT IN (4,5)

Options:

A. B, A, A

B. A, A, B

C. C, B, C

D. A, C, A

Q6 All employees with department names

Part 1

A. SELECT e.name, d.dept_name

B. SELECT e.name

C. SELECT dept_name

Part 2

A. employees e INNER JOIN departments d

B. employees e LEFT JOIN departments d

C. departments RIGHT JOIN employees

Part 3

A. ON e.id = d.id

B. ON id = dept_id

C. ON e.dept_id = d.id

Options:

A. B, A, A

B. A, B, C

C. C, C, B

D. A, A, B

Q7 Assign row numbers by salary

Part 1

A. SELECT name, ROW_NUMBER() OVER (ORDER BY salary DESC) AS rn

B. SELECT salary

C. SELECT name, RANK()

Part 2

A. employees

B. departments

C. employee_data

Part 3

A. WHERE salary > 1000

B. ORDER BY salary

C. GROUP BY salary

Options:

A. B, A, B

B. A, A, A

C. C, A, C

D. A, B, A

Q8 Departments with avg salary > 50000

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Part 1
A. SELECT dept_id
B. SELECT dept_id, AVG(salary)
C. SELECT AVG(salary)

Part 2
A. employees
B. departments
C. salaries

Part 3
A. WHERE salary > 50000
B. GROUP BY dept_id
C. GROUP BY dept_id HAVING AVG(salary) > 50000

Options:
A. B, A, C
B. A, A, A
C. C, B, B
D. B, C, A

Q9 Employees in large departments

Part 1
A. SELECT name
B. SELECT salary
C. SELECT dept_id

Part 2
A. employees
B. departments
C. employee_data

Part 3
A. WHERE salary IN (1000,2000)
B. WHERE dept_id IN (SELECT dept_id FROM employees GROUP BY dept_id HAVING COUNT(*) > 10)
C. WHERE dept_id > 10

Options:
A. B, A, A
B. A, A, B
C. C, B, C
D. A, C, A

Q10 Rank employees by salary

Part 1
A. SELECT name, RANK() OVER (ORDER BY salary DESC) AS rnk
B. SELECT salary
C. SELECT name, COUNT(*)

Part 2
A. employees
B. departments
C. employee_data

Part 3
A. WHERE salary > 0
B. ORDER BY salary
C. GROUP BY salary

Options:
A. B, A, B
B. A, A, A
C. C, A, C
D. A, B, A

In Data Analytics, the data is the raw stone, but **SQL is your chisel**.

Don't worry if your query fails on the first try; even the best Data Analysts have been humbled by a missing comma. **Good luck!**

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